

Tran Quoc Truong

Applied AI Backend Engineer | Agent Evaluation & RAG Systems

📍 Ho Chi Minh City, Vietnam ✉ tranquoctruong20@gmail.com 🌐 truong51972.github.io/about-me
🌐 [truong51972](#) 🐙 [truong51972](#)

Summary

Applied AI Backend Engineer focused on building AI agent workflows, RAG systems, and agent evaluation platforms using Python, Django/DRF, FastAPI, async workers, and vector databases.

Built ATA, an AI-native validation and audit platform for agentic systems, combining traceable model-provider capture, LLM evaluation, quality gates, and production-oriented backend workflows.

Experience

FPT Software, Automation Tester & Applied AI Engineer

Ho Chi Minh City, Vietnam

Dec 2024 – present

1 year 8 months

- Built applied AI workflows for document understanding, RAG Q&A, test-case generation, duplicate bug detection, traceability, and change impact analysis.
- Designed Python backend orchestration with Django/DRF, FastAPI, Redis/Celery, PostgreSQL, async workers, service layers, and clear domain boundaries.
- Developed LLM evaluation, RAG validation, and agent-quality patterns using judge/critic/meta-judge workflows, evidence capture, and regression-oriented reporting.
- Applied banking QA domain experience to AI quality systems, preserving auditability, transaction integrity, and blocker-based validation in backend workflows.

FPT Software Academy, Full-Stack Developer Intern

Ho Chi Minh City, Vietnam

Sept 2024 – Dec 2024

4 months

- Built Django-based web applications and deployed containerized services with Docker.
- Improved deployment reliability through production-oriented configuration and Cloudflare integration.
- Reduced deployment cycle time by approximately 50% through cleaner containerized delivery workflows.

Projects

Omni-Agent — Django/DRF AI Agent Hub for Knowledge Builder & Agent Testing

A Django/DRF-based AI Agent Hub where the main backend owns product logic, domain state, workflow rules, jobs, artifacts, and APIs, while stateless workers and services execute technical AI tasks.

- Designed Omni-Agent as the main Django/DRF backend for project-centric AI agent products, owning domain logic, permissions, workflow state, job lifecycle, artifact metadata, and product module boundaries.
- Built Knowledge Builder as the first product module, supporting versioned document workflows, OCR/chunking pipelines, S3-compatible artifacts, presigned preview/download URLs, and traceable RAG foundations.
- Modeled external workers and REST services as stateless executors for technical tasks such as OCR, document parsing, chunking, embedding, retrieval, and evaluation, with Django persisting domain results.
- Positioned ATA as a completed standalone validation platform that can become Omni-Agent's Agent Testing and Audit module for evaluating RAG and LLM-agent workflows.
- Prototyped React/Vite console workflows for project navigation, product switching, document workspace review, version selection, preview tabs, and processing status visibility.

ATA — Agent Test Agent

Independent validation and audit layer for AI agent systems that captures model-provider interactions, scores records across 10 dimensions, and produces PASS/WARN/FAIL/BLOCK workflow verdicts.

- Built a completed standalone MVP with a FastAPI control plane, LiteLLM capture gateway, Celery validation worker, and React/Vite dashboard.
- Used shared SQLAlchemy ORM, domain contracts, PostgreSQL, and Redis to keep API, gateway, worker, and dashboard state consistent across services.
- Implemented a Planner/Judge/Critic/Meta Judge validator that scores records across context, tools, reasoning, security, compliance, operations, human validation, and business outcome dimensions.
- Designed a distribution-based Workflow Quality Gate using P10/P35/P50, mean, deviation, and blocker dimensions to prevent weak low-tail records from being hidden by averages.
- Enforced fail-closed credential handling with ATA-prefixed recording keys and request-local provider secrets that are never persisted to Redis, PostgreSQL, logs, or UI state.

APIT — Agent Programmatic Integration Testing

An LLM-based system that analyzes API documentation and generates structured API test cases.

- Built the core AI agent for API document understanding, structured test-case generation, and automated output evaluation.
- Evaluated LoRA and QLoRA approaches for domain-specific structured test document generation.
- Fine-tuned Qwen-2.5-3B, achieving macro-F1 0.655 and outperforming Llama-3.2-3B by 41.5% in the evaluation setup.
- Implemented FastAPI-based model evaluation workflows and LLM-as-a-Judge experiments for generated output comparison.

Skills

Applied AI & Agent Evaluation: RAG, Hybrid RAG, LangGraph, LangChain, LiteLLM, Langfuse, LLM Evaluation, LLM-as-a-Judge, eval datasets, judge/critic/meta-judge patterns

Backend & Systems: Python, Django/DRF, FastAPI, PostgreSQL, Redis/Celery, Docker, REST APIs, async workers, service layers, domain modeling

Agent Evaluation & AI Quality: Agent testing, AI quality gates, evidence capture, API validation, SQL validation, regression analysis, Core Banking, Payment Hub

Frontend Prototyping: React, Vite, Tailwind CSS, internal dashboard UI, AI-assisted development

Education

FPT School of Business & Technology, Master of Software Engineering in AI

Ho Chi Minh City, Vietnam
May 2026 – present

FPT University, Bachelor of Artificial Intelligence

Ho Chi Minh City, Vietnam
2021 – 2025

Languages

Vietnamese: Native

English: Working proficiency